Discrete Structures I

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What we are going to Cover for Topic III

- Using sets as our basic object, and guided by the logic we have established since Section 1, this chapter will build up more complex objects.
- Although more complex, these so-called basic structures are still very “basic” and fundamental to the mathematical foundations of computer science.
- In this chapter we will explore functions, sequences, sums, and series

A Formal Definition

Definition 1.1: Function

A function f is a binary relation from a set X to a set Y such that for every $x \in X$, there is exactly one $y \in Y$ such that $(x, y) \in f$

- Notice that the definition of a function is as a special kind of binary relation.
- We have previously seen reflexivity, symmetry, transitivity, and antisymmetry.
- Now, we see that a function is a special kind of relation with two properties:
  1. Every element of X must appear as the first coordinate of some pair in the binary relation f.
  2. Every element of X must appear only once in the binary relation f.

  When these two properties hold, a binary relation is a function

Functions

- The terms function and map are often used interchangeably.

Example 1.1: Binary Relations and Function

Consider the following binary relations from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$. Which of the following binary relations are functions?

$$\mathcal{R}_1 = \{(1, a), (2, b), (3, c)\}$$
$$\mathcal{R}_2 = \{(1, a), (2, a), (3, a)\}$$
$$\mathcal{R}_3 = \{(1, a), (1, b), (2, b), (3, c)\}$$
$$\mathcal{R}_4 = \{(1, a), (1, b), (2, b), (2, c)\}$$
$$\mathcal{R}_5 = \{(1, c), (2, a)\}$$

Solution

Functions

- The terms function and map are often used interchangeably.

Example 1.2: Binary Relations and Function

Consider the following binary relations from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$. Which of the following binary relations are functions?

$$\begin{aligned}
\mathcal{R}_1 &= \{(1, a), (2, b), (3, c)\} \\
\mathcal{R}_2 &= \{(1, a), (2, a), (3, a)\} \\
\mathcal{R}_3 &= \{(1, a), (1, b), (2, b), (3, c)\} \\
\mathcal{R}_4 &= \{(1, a), (1, b), (2, b), (2, c)\} \\
\mathcal{R}_5 &= \{(1, c), (2, a)\}
\end{aligned}$$

Solution

1. $\mathcal{R}_1$ is a function
2. $\mathcal{R}_2$ is a function
3. $\mathcal{R}_3$: Not a function: 1 appears twice
4. $\mathcal{R}_4$: Not a function,3 does not appear in any pair and 1 and 2 both appear twice
5. $\mathcal{R}_5$: Not a function,3 does not appear in any pair

Domain & Codomain

- Special terminologies are used where binary relation is a function.
- Let $\mathcal{R}$ be a binary relation from the set A to the set B. If $\mathcal{R}$ is a function, lower-case roman letters: f, g, h etc to signify a function
- The notation $f : A \to B$ signify that the function f is from set A to set B
- We say f maps A to B. A is called the domain of the function and B is the codomain of the function

Definition 1.2: Domain

When a binary relation from set A to set B is a function, we call A the domain of the function.

Definition 1.3: Codomain

When a binary relation from set A to set B is a function, we call B the codomain of the function.

Image and Preimage

- Let $f : A \to B$ be a function. We write $f(a) = b$ where $a \in A, b \in B$ and $(a, b) \in f$
- $f(a)$ is called the *image* of a under f while a is called the *preimage*

Construction Functions

- Functions can be constructed and represented in many ways. Indeed, functions are binary relations and binary relations are sets.
  1. Discrete functions using roster methods
  2. Functions as formulas or Functional and Arrow Notations
  3. Functions as Algorithms
  4. Functions as graphs

1. Discrete functions using roster methods

- Roster method, just as in sets, is one of the representation of function.
- Given a finite domain, it is possible to simply list all of the pairs in the function.

Example 1.3: A discrete and finite function

Let S be the set of students $\{\text{Andrea}, \text{Bruce}, \text{Davood}, \text{Fiona}, \text{Jalal}, \text{Rahul}\}$.

Let M be the set of majors
$\{\text{Computer Science}, \text{Mathematics}, \text{Physics}, \text{Chemistry}, \text{Biology}\}$

A function $f : S \to M$ maps each student in S to a unique element of M

$$
\begin{aligned}
f = \{ & \\
& (\text{Andrea, Computer Science}), \\
& (\text{Bruce, Biology}), \\
& (\text{Davood, Physics}), \\
& (\text{Fiona, Computer Science}), \\
& (\text{Jalal, Physics}), \\
& (\text{Rahul, Chemistry}) \\
\} &
\end{aligned}
$$

2. Functions as formulas

- A more simple description of a function is a formula
- Consider a simple linear function f.
  Thre are two ways to express it:

$$f(x) = 2x + 5$$

$$x \mapsto 2x + 5$$

- The former is common in calculus. It is called functional notation while the arrow notation is used to describe without giving a particular name.
- In both cases, the domain and codomain can be implicit by context, and typically are both $\mathbb{R}$

2. Functions as formulas

Example 1.4: Functional and Arrow Notation

The following two notations define the same real quadratic function.

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = 2x^2 - 5x + 3$$

$$f : \begin{array}{l} \mathbb{R} \to \mathbb{R} \\ x \mapsto 2x^2 - 5x + 3 \end{array}$$

3. Functions as algorithms

- Functions can also be represented as algorithms
- Note that this does not mean that all computer functions or computer algorithms are functions. Indeed, void computer functions are not mathematical functions.
- Moreover, *methods*, which operate on objects in object-oriented languages are typically not mathematical functions because they use some notion of “state”.
- An algorithm which returns a value solely depending on its input arguments (i.e. has no state), can represent a mathematical function.

4. Functions as graphs.

- So far, we have not yet defined formally what a graph is.
- However, we can still understand and represent functions pictorially using Venn diagrams, nodes, and arrows for at least, when the function is discrete and finite.
- The image below represents the function $\{(a, 2), (b, 3), (c, 1), (d, 3)\}$

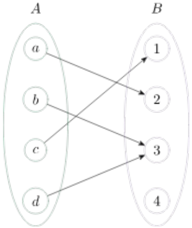

Figure 1: A function from $A = \{a, b, c, d\}$ to $B = \{1, 2, 3, 4\}$

Function Range

Definition 1.4: Range

The range of a function $f : A \to B$ is the set of all images of f under A. It is denoted by $f(A)$

- The range of a function is a (not necessarily strict) subset of its codomain. As we will see later, functions have special properties when their range equals their codomain.
- For a function $f : A \to B$ we can extend the notation of its range $f(A)$ to any subset of A. For $S \subseteq A$ we have

$$f(S) = \{f(s) \mid s \in S\}.$$

Properties of Functions

Example 1.5: Properties of functions

Consider the following function $f : A \to B$ described by the figure 2

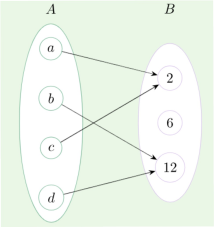

Figure 2: Properties of functions

What is
(i) $f(a)$?
(ii) The image of d?
(iii) The domain of f?
(iv) The range of f
(v) The preimage of 12?
(vi) $f(A)$?
(vii) The preimage of 6?
(viii) $f(\{a, c, d\})$?

Example: Properties of functions - Solution Cont'd

i. 2
ii. 12
iii. $\{a, b, c, d\} = A$
iv. $f(A) = \{2, 12\}$
v. $\{b, d\}$
vi. There is none
vii. $\{2, 12\}$

Partial Functions

- A binary relation is a function if it is total and univalent

Definition 1.5: Total

A binary relation $\mathcal{R}$ from A to B is total if, for every element $a \in A$ there exists an element $b \in B$ such that $(a, b) \in \mathcal{R}$

Definition 1.6: Univalent

A binary relation $\mathcal{R}$ from A to B is univalent or right-unique if, for every $a \in A$ and for every $b, c \in B$ we have

$$(a, b) \in \mathcal{R} \wedge (a, c) \in \mathcal{R} \rightarrow b = c$$

- A binary function that is univalent but not total is a partial function.

Partial Functions

- For a partial function, the set of all preimages is a subset of the domain and is called the domain of definition.
- Let $f : A \rightarrow B$ be a partial function.
- Let $S \subseteq A$ be its domain of definition.
- We say that the partial function is undefined for element in $A \setminus S$

Example 1.6: Square Root as a Partial Function

Consider the function $f : \mathbb{N} \rightarrow \mathbb{R}$ where $f(n) = \sqrt{n}$. This is a partial function from $\mathbb{Z}$ to $\mathbb{R}$ since the square root of negative integers is undefined when the codomain is the real numbers

Partial Functions

- Example 1.6 raises an important property of functions that a formula alone does not define a function.

Example 1.7

Consider $f(n) = \sqrt{n}$ from $\mathbb{R}$ to $\mathbb{C}$. In this case the image of negative numbers under f are all well defined complex numbers

Injective, Surjective, Bijective Functions

- In addition to total and univalent, we may strengthen the conditions on a function to get particular kinds of very useful functions.
- Functions that are one-to-one, onto, or invertible are very important in practice.

Definition 1.7

A function is injective if every element in the range of the function has a unique preimage. That is, for a function $f : A \to B, \forall a_1, a_2 \in A, a_1 \neq a_2 \to f(a_1) \neq f(a_2)$

Injective, Surjective, Bijective Functions

- When a function is injective, we call it an injection or a one-to-one function.

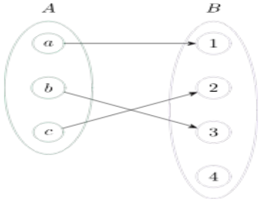

Figure 3: An injective function from A to B

- One way to remember what “injective” means is to think “into”.
- Injective functions transform or map elements of A into elements of B.
- That is, the transformed elements of A can be uniquely found within B.

Injective, Surjective, Bijective Functions

Definition 1.8: Surjective

A function is surjective if its codomain and range are equal. That is, for a function $\forall\, b \in B \exists\, a \in A$ such that $f(a) = b$

- When a function is surjective, we call it a surjection or an onto function.

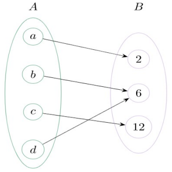

Figure 4: A surjective function from A to B

- One way to remember what “surjective” means is to think “onto”.
- From French, we can think “sur” means “on”.
- From Latin, “sur” also means “over”, “above”, “on top”.
- A surjective function fully “covers”, “shadows” or is “on top of” its codomain.

Injective, Surjective, Bijective Functions

- For an surjective function $f : A \to B$ we must have $|A| \geq |B|$

Definition 1.9: Bijective

A function is bijective if it is both injective and surjective.

- When a function is bijective, we call it a bijection or an invertible function or a one-to-one correspondence.
- Note that the “correspondence” is very important.
- A one-to-one function may not be a bijection and therefore may not be a one-to-one correspondence.

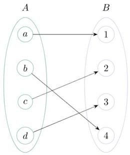

Figure 5: A bijective function from A to B

Injective, Surjective, Bijective Functions

- Notice that a bijective function can always be rearranged so that:
  1. The mappings are all parallel to each other, figure 5 is the same as figure 6
  2. Every element of the domain maps to a unique element of the codomain, and
  3. Every element of the codomain has a unique premiage.

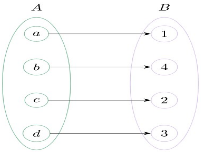

Figure 6: A bijective function from A to B

Injective, Surjective, Bijective Functions

Example 1.8: Injective, Surjective, Bijective Functions

For the following functions, determine if they are surjective, injective, or bijective. If so, give a proof. If not, supply a counterexample.

Suppose $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4, 5\}$

$$f_1 = \{(a, 1), (c, 4), (b, 3), (d, 2)\}$$

$$f_2 : \begin{array}{l} \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \\ (n, m) \mapsto 3n + 2m \end{array}$$

Solution

Injective, Surjective, Bijective Functions

Example 1.9: Injective, Surjective, Bijective Functions

For the following functions, determine if they are surjective, injective, or bijective. If so, give a proof. If not, supply a counterexample.

Suppose $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4, 5\}$

$$f_1 = \{(a, 1), (c, 4), (b, 3), (d, 2)\}$$

$$f_2 : \begin{array}{l} \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \\ (n, m) \mapsto 3n + 2m \end{array}$$

Solution

f_1 is not a surjection because 5 does not have a preimage. f_1 is an injection since every element of A has a unique image in B.

f_2 is not injective since, for example, $f(3, 4) = 9 + 8 = f(1, 7) = 3 + 14 = 17$. f_2 is surjective, since for any integer x we can solve $x = 3n + 2m$ using $(x, -x)$

Special Funictions

- There are certain functions which are quite important and we give them special names.
- For example, "quadratic functions" and "cubic functions" are polynomial functions of degree 2 and 3, respectively.
- These names describe an entire set of functions.
- Indeed, there are infinitely many quadratic functions over the reals.
- They are described by the set:

$$\left\{ f : \begin{array}{l} \mathbb{R} \to \mathbb{R} \\ x \mapsto ax^2 + bx + c \end{array} \;\middle|\; a, b, c \in \mathbb{R} \right\}$$

Special Functions: 1. Identity Function

- The identity function is a function from a set to the same set which always has its value equal to its argument. That is, $f(a) = a$ for all a in the domain of f
- Given a set A, we denote the identity function on the set A as id_A. We have

$$\mathrm{id}_A : \begin{array}{l} A \to A \\ x \mapsto x \end{array}$$

- Note that the identity function on a set A is the unique bijective and reflexive function on A.

Example 1.10: A Discrete Idenity Function

Let $A = \{1, 2, 3, 4, 5\}$. The identity function on A is

$$\mathrm{id}_A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$$

Special Functions: 2. Constant Function

- A constant function is a function whose value is the same for every possible input.
- A constant function can map from one set to another, or from one set to itself.
- The only requirement is that the value of the function is always exactly the same value regardless of the input.

Example 1.11: A constant and discrete example

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. The following are all different constant functions from A to B

$$f = \{(a, 1), (b, 1), (c, 1)\}$$

$$g = \{(a, 2), (b, 2), (c, 2)\}$$

$$h = \{(a, 3), (b, 3), (c, 3)\}$$

Special Functions: 3. Floor & Ceiling

- Two useful functions whose codomain are the integers are called ceiling and floor.
- We can think of ceiling as "rounding up" a number to the nearest integer while floor is "rounding down" to the nearest integer.
- For a number x we denote its ceiling as $\lceil x \rceil$ and it floor as $\lfloor x \rfloor$
- Formally, ceiling is a function from $\mathbb{R}$ to $\mathbb{Z}$ defined as

$$\lceil x \rceil = z \in \mathbb{Z} \mid z - 1 < x \leq z$$

- Meanwhile, floor is a function from $\mathbb{R}$ to $\mathbb{Z}$ defined as

$$\lfloor x \rfloor = z \in \mathbb{Z} \mid z \leq x < z + 1$$

Special Functions: 3. Floor & Ceiling

Example 1.12: Hitting the floor

- We have the following equalities.
  1. $\lfloor 4.5 \rfloor = 4$
  2. $\lfloor 3.999 \rfloor = 3$
  3. $\lceil 4.0001 \rceil = 5$
  4. $\lceil \sqrt{3} \rceil = 2$
  5. $\lfloor -4.3 \rfloor = -5$
  6. $\lceil -10.9 \rceil = -10$

Special Functions: 4. Factorial Functions

- The last important function which we will explore in this topic is the factorial function.
- It will become extremely useful in our study of combinatorics.
- The factorial function is a function $f : \mathbb{N} \to \mathbb{Z}^+$ defined by $f(n) = n!$ with:

$$n! = \begin{cases} 1, & n = 0 \\ 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n, & \text{otherwise} \end{cases}$$

- The factorial function is known for growing extremely quickly.
- In particular, its values grow faster than exponential growth.
- For relatively small arguments, the factorial function has very very large values.

Inverse Functions

- An invertible function is one which has an inverse. What is an inverse?

Definition 1.10: Inverse Function

Given a bijection f from set A to set B then the inverse of f, denoted f^1 is a function from B to A defined as

$$f^{-1}(b) = a \leftrightarrow f(a) = b$$

- Given a function, its inverse is a function that “reverses” the mapping caused by the original function.
- The inverse function “undoes” the transformation caused by the original function.
- It should be clear from the definition of an injective function that only injective functions are reversible in this way.

Inverse Functions

- Indeed, each image needs a unique preimage in order to reverse the original mapping.
- However, we additionally require that invertible functions are surjective.
- Remember that a defining feature of functions is that they are total over their domain.
- If a function is not surjective, then its reverse mapping will not be total and thus not a function.
- These facts tell us that invertible functions are necessarily bijective functions, and, bijections necessarily have inverses.

Inverse Functions

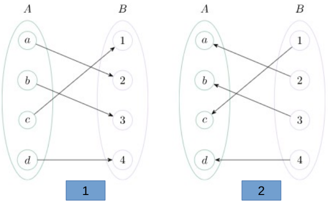

Figure 7: Number 1 is a function f from A to B whereas number 2 is the inverse f from B to A

Inverse Functions

Example 1.13: Inverse Example

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \frac{1}{2}x + 3$$

This linear function clearly has an inverse. Indeed, we can "solve" for $f(x)$. Letting $y = f(x)$

$$\begin{aligned} f(x) =& \frac{1}{2}x + 3 \\ y =& \frac{1}{2}x + 3 \\ y - 3 =& \frac{1}{2}x \\ 2y - 6 =& x \end{aligned}$$

$$\to f^{-1}(y) = 2y - 6$$

$g \circ f$:

- The composition of functions is an operation performed on functions themselves (not elements of their domains), to produce another function.

Definition 1.11: Composition

Given a function $f : A \to B$ and a function $g : B \to C$, the composition of g and f is the function $g \circ f : A \to C$ defined as $(g \circ f)(a) = g(f(a))$

Funciton Composition

- The composition of two functions takes the range of the second function and uses it as the domain of the first.
- That is, the "output" of the second is used as the "input" to the first. In this way, function composition "chains" functions together.

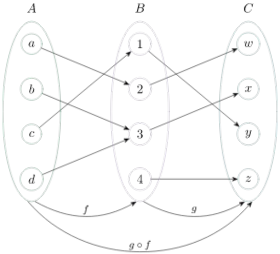

Figure 8: The composition of functions $f : A \to B$ and $g : B \to C$

- In fig. 8 $f(c) = 1, g(1) = y$ and $(g \circ f)(c) = g(f(c)) = y$
- Notice also that g is a bijection. However $f \circ g$ is not surjective.

Funciton Composition

Example 1.14: Order of Composition

Let $f(x) = x^2$ and $g(x) = 2x + 1$ be two functions over the reals

$$(f \circ g)(x) = f(g(x)) = (2x + 1)^2$$

$$(g \circ f)(x) = g(f(x)) = 2x^2 + 1$$

- Lastly, a discussion on function compositions would be incomplete without considering again function inverses.
- Function inverses have a great property that a function composed with its inverse gives the identity function.
- For example, Consider any invertible function over a set $A. f : \mathbb{A} \rightarrow \mathbb{A}$ It holds that:

$$f \circ f^{-1} = f^{-1} \circ f = \mathrm{id}_A$$

Proofs with Functions

- Proofs with functions follow similarly to proofs over predicate logic or proofs with sets.
- They begin by appealing to the definition of the object in question and then applying one of our many proof techniques: direct, contrapositive, contradiction, cases, etc.
- The kinds of proofs regarding functions which we will do in this course applies to discrete functions and properties which can be proved without the need for analysis. But, some basic calculus (like limits) may be helpful.

Proofs with Functions

Theorem 1.1: Composing Bijections

If $f : A \to B$ and $g : B \to C$ are invertible functions, the $(g \circ f)$ is invertible.

Proof.

- Assume f and g are bijections. $(g \circ f)$ is a function from A to C. Let $x \in A$. By definition, we have that $(g \circ f)(x) = g(f(x))$
- Let us prove by surjection. Since f is a bijection, $f(A) = B$. Since g is a bijection, $g(B) = C$. Hence $(g \circ f)(A) = g(f(A)) = C$ and the composition is surjective.
- We now prove that the composition is injective. Let $x_1, x_2 \in X$. We will show that $(g \circ f)(x_1) = (g \circ f)(x_2) \to x_1 = x_2$. Since g is a bijection, $g(f(x_1)) = g(f(x_2)) \to f(x_1) = f(x_2)$
- Since f is a bijection, $f(x_1) = f(x_2) \to x_1 = x_2$. Therefore, $(g \circ f)$

□

Proofs with Functions

- Another useful property of functions that will helps in proofs is monotonicity. This is described by the following definitions.

Definition 1.12: Non-Decreasing

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is non-decreasing if for all $x, y \in \mathbb{R}, x \leq y \rightarrow f(x) \leq f(y)$. A strictly non-decreasing or strictly increasing function has $x < y \rightarrow f(x) < f(y)$

Definition 1.13: Non-Increasing

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is non-increasing if for all $x, y \in \mathbb{R}, x \leq y \rightarrow f(x) \geq f(y)$. A strictly non-increasing or strictly decreasing function has $x < y \rightarrow f(x) > f(y)$

Proofs with Functions

Definition 1.14: Monotonic

A function is monotonic if it is either non-increasing or non-decreasing. A function is strictly monotonic if it is either strictly non-increasing or strictly non-decreasing.

- The graph of monotonic functions do not have turning points, points where the graph goes from moving upwards, to moving downwards, or vice versa.
- Turning points are special kinds of stationary points, where the derivative of a function is 0

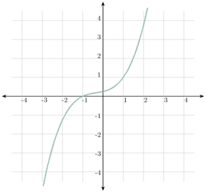

Figure 9: The cubic $f(x) = \frac{1}{4}(x^3 + x^2 + x + 1)$ is strictly increasing

Monotonic

- We can verify that the function shown in the graph above is monotonic by computing its derivative $\frac{d}{dx}f(x) = \frac{1}{4}(3x^2 + 2x + 1)$
- This derivative is always positive and hence the function is strictly increasing.
- A useful property of strictly monotonic functions is that they are always injective.

Proposition 1.1

Any function f which is strictly monotonic is injective.

Proof.

- We look to show that "f is strictly monotonic" $\rightarrow f$ is injective
- A function is strictly monotonic if it is either strictly increasing or strictly decreasing, yielding two cases.

Monotonic

Proof Cont'd.

1. strictly increasing. By definition, we have
   $\forall x, y\ x < y \to f(x) < f(y)$
   Therefore, for any $x_1 \neq x_2$ we have $f(x_1) \neq f(x_2)$ since either $x_1 < x_2$ and $f(x_1) < f(x_2)$ or $x_1 > x_2$ and $f(x_1) > f(x_2)$
2. strictly decreasing. By definition, we have
   $\forall x, y\ x < y \to f(x) > f(y)$. Therefore, for any $x_1 \neq x_2$ we have $f(x_1) \neq f(x_2)$ since either $x_1 < x_2$ and $f(x_1) > f(x_2)$ or $x_1 > x_2$ and $f(x_1) < f(x_2)$

□

Exercises

1. Let $A = \{1, 2, 3, 4\}$ Which of the following binary relations defines a function from the set A? If not, explain why
   a.) $f = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$
   b.) $f = \{(1, 2), (2, 3), (3, 4)\}$
   c.) $f = \{(4, 2), (2, 4), (3, 4), (4, 2)\}$
   d.) $f = \{(1, 4), (3, 2), (2, 3), (1, 3), (4, 2)\}$
   e.) $f = \{(4, 4), (1, 4), (2, 4), (3, 4)\}$
   f.) $f = \{(1, 8), (2, 1), (3, 9), (4, 6)\}$
2. Consider the set S of students currently enrolled at courses at JKUAT, the set P of Professors of JKUAT and the set C of course offerings in the current semester. Under what circumstances would each of the following relations actually a function?
   a.) $\{(s, p) \in S \times P \mid s \text{ is enrolled in a course taught by } p\}$
   b.) $\{(p, s) \in P \times S \mid p \text{ teaches } s\}$
   c.) $\{(p, c) \in P \times C \mid p \text{ teaches } c\}$
   d.) $\{(c, p) \in C \times P \mid c \text{ is taught by } p\}$
3. Let $E = \{3x \mid x \in \mathbb{N}\}$. Prove that E is countable

Exercises

4. Let $S = \{1, 2, 3, 4, 5\}$. Let $f : S \to \mathbb{Z}$ be a function defined as

$$f(x) = \begin{cases} 2x - 6, & x \text{ is even} \\ x^2 + 2 & x \text{ is odd} \end{cases}$$

Describe f as a binary relation using the roster method. Show that f is one-to-one

5. For each of the following functions, determine its range. Is the function an injection, surjection, or bijection? If so, explain why (not necessarily a formal proof). If not, give a counterexample.

$$f_1 : \begin{cases} \mathbb{R} \to \mathbb{R} \\ x \mapsto x^3 + 5x^2 \end{cases}$$

$$f_2 : \begin{cases} \mathbb{R} \to \mathbb{R} \\ x \mapsto x^4 + 2x^2 + 10 \end{cases}$$

$$f_3 : \begin{cases} \mathbb{R} \to \mathbb{R}^+ \\ x \mapsto e^x \end{cases}$$

Exercises

6. For each of the following functions, determine if they are a injection, surjection, or bijection. Explain why.
   (i) $f_1 : \mathbb{Q} \rightarrow \mathbb{Q}$, $f(x) = 3x + 2$
   (ii) $f_2 : \mathbb{N} \rightarrow \mathbb{N}$, $f(x) = 3x + 2$
   (iii) $f_3 : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, $f(x, y) = 5x + 3y$
   (iv) $f_4 : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x, y) = 5x + 3y$
7. Consider the function $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ such that $f(x) = \frac{1}{x}$. Prove that f s a bijection. Recall that $\mathbb{Q}^+ = \{a \in Q \mid a > 0\}$

Introduction

- Sequences are nothing more than ordered lists of elements.
- We can feel that they are strongly related to tuples.
- However, tuples are always finite.
- Sequences, on the other hand, may be infinite.

Sequences

- Informally and often in practice, a sequence is nothing more than a list of elements:

$$1, 2, 3, 4, 5, 6$$

$$a, f, c, e, g, w, z, y$$

$$1, 1, 2, 3, 5, 8, 13,$$

$$0, 1, 0, 1, 0, 1, 0, 1, 0, 1, \ldots$$

- A sequence can be finite or finite. Elements of a sequence can be repeated.
- Formally speaking, sequences are functions from a subset of the integers to another set S

Sequences

Definition 2.1: Sequence

A sequence is a function from the subset of the integers, typically $\mathbb{N}$ to a set S

- When referring to a sequence, we use the notation a_n to refer to the image of n under the sequence function.
- So, $f(n) = a_n$ for some sequence defined by f.
- When speaking of functions which encode sequences, a value of the function is instead called a term of the sequence.
- This definition is very generic. It imposes no properties on the function.
- For example, the function does not need to be surjective nor injective.
- The domain and codomain of the function are also left quite generic.

Sequences Representation

1. Roster-like method: This case writes all the terms of the sequence, leaving the domain of the function encoding the sequence as implicit. Often the domain is either $\mathbb{N}$ or $\mathbb{N} \setminus \{0\}$ depending on the conventions and context.
   Note that the domain of sequences must be integers

Example 2.1

The sequence

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots,$$

is the decreasing sequence of positive fractions with 1 as the numerator. The first term is clearly 1 labelled $a_1 = 1$. This makes the sequence defined formally as

$$f : \begin{cases} \mathbb{N} \setminus \{0\} \to \mathbb{Q} \\ n \mapsto \frac{1}{n} \end{cases}$$

If the first element is a_0, then this set would be defined formally as

$$f : \begin{cases} \mathbb{N} \to \mathbb{Q} \\ n \mapsto \frac{1}{n+1} \end{cases}$$

Sequences Representation

2. Using a Formula and domain indicator. In this case, we us the roaster method but instead of writing the actual terms, we write $a_1, a_2, \dots$ next to the formula
   $a_n = \frac{1}{n+1}$ $\qquad (a_n) = (a_0, a_1, a_2, a_3, \dots)$.
   Parenthesis are used to denote sequences, much like tuples
3. Using the index range. The index range defines the range of values that the index i may take for the terms a_i

$$(a_n) = \frac{1}{n+1} \quad \text{for } i \geq 0$$

or simply

$$(\frac{1}{n+1}) \quad \text{for } i \geq 0.$$

An even further shorthand would be

$$(\frac{1}{n+1})_{n \in \mathbb{N}}$$

Progressions

- There are kinds of sequences to which we give special names. We will look at two:
  1. arithmetic progressions
  2. geometric progressions.

Arithemtic Progression.

- An arithmetic progression is a sequence with a common difference between each term.
- Given an initial term a and a common difference d, an arithmetic progressions is a sequence of the form:

$$(a, a+d, a+2d, a+3d, a+4d, \ldots)$$

- Therefore, $a_n = a + nd$ for $n \in \mathbb{N}$
- In an arithmetic progression, the key trait is that $a_{i+1} - a_i = d$. The starting term and the difference can be any (typically real) number.
- Notice that an arithmetic progression can either be increasing or decreasing, depending on if d is positive or negative.

Geometric Progression.

- Geometric progressions are sequences with a common ratio between each term.
- Given an initial term a and a common ratio r, an arithmetic progression is a sequence of the form:

$$(a, ar, ar^2, ar^3, ar^4, \dots)$$

- Therefore, $a_n = ar^n$ for $n \in \mathbb{N}$
- In a geometric progression, the key trait is that $a_{i+1}/a_i = r$ The starting term and ratio can be any (typically real) number.
- A geometric progression may encode exponential growth or exponential decay depending on r
- If $r > 1$, then it is exponential growth.
- If $0 < r < 1$ then it is exponential decay.
- Another fun property of geometric progressions are that if r is negative, then the sequences goes back and forth between positive and negative numbers.

Recurrence Relations

- Recurrence relations are descriptions for special kinds of sequences where the definition of one term relies on the previous term(s).

Definition 2.2: Recurrence Relation

A recurrence relation for a sequence (a_n) is an equation defining a_n in terms of one or more of the previous terms in the sequence

- A recurrence relation should be paired with initial conditions.
- A sequence which satisfies the formula of a recurrence relation is called a solution of the recurrence.
- Arithmetic progressions and geometric progressions can be defined using recurrence relations.

Solving a recurrence relation

- Recall that a solution to a recurrence relation is a sequence of terms satisfying the relation.
- In order to solve a recurrence relation we must determine a closed-form formula for a_n which depends on n and does not depend on any previous term
- Solving recurrence relations is a crucial part of computer science.
- Indeed, the efficiency of an algorithm to compute an element of a sequence is (most often) greatly improved by a closed-form formula rather than a recursive algorithm.
- There are several strategies to try and solve recurrence relations. Two common strategies are
  1. forward substitution and
  2. backward substitution

Solving a recurrence relation

Example 2.2: Forward substitution

- Solving a recurrence relation $a_n = a_{n-1} + 3$ with $a_0 = 2$
- Forward substitution starts at the initial condition and applies the recurrence relation over and over.

$$\begin{aligned}
a_0 &= 2 \\
a_1 &= 2 + 3 \\
a_2 &= (2 + 3) + 3 = 2 + 2 \cdot 3 \\
a_3 &= (2 + 2 \cdot 3) + 3) = 2 + 3 \cdot 3 \\
&\vdots \\
a_n &= a_{n-1} + 3 = (2 + (n-1)3) + 3 = 2 + 3n
\end{aligned}$$

Solving a recurrence relation

Example: Backward substitution

- Backward substitution starts at the recurrence relation and attempts to backtrack to the initial condition. Using the same example

$$
\begin{aligned}
a_n &= a_{n-1} + 3 \\
&= (a_{n-2} + 3) + 3 = a_{n-2} + 2 \cdot 3 \\
&= (a_{n-3} + 3) + 2 \cdot 3 = a_{n-3} + 3 \cdot 3 \\
&= \vdots \\
&= (a_1 + 3) + (n-2)3 = a_1 + (n-1)3 \\
&= (a_0 + 3) + (n-1)3 = a_0 + 3n = 2 + 3n
\end{aligned}
$$

Solving a recurrence relation

Solving recurrence relations is typically a very challenging process. Many recurrence relations simply cannot be solved.

Example 2.3: Complete Compounding interest Rec. Rel

Consider a savings account which has an annual interest rate of 3%, compounding annually. If an investor makes an initial deposit of KES 1000.00, how much money will be in the account after 25 years?

Series

- Before we look at the more general object of a series, consider first a summation.
- A summation is simply a sum of a collection of numbers.
- Because addition is commutative, the order in which numbers of the collection are added does not matter.
- However, for ease of description and notation, we define a summation as a the addition of all the terms in a finite sequence.
- The resulting value is the sum.
- Generally, for a sequence a_i for $c \leq i \leq n$ we write its summation using summation notation:

$$\sum_{i=c}^{n} a_i = a_c + a_{c+1} + a_{c+2} + \cdots + a_n, \quad \text{or}$$

$$\sum_{c \leq i \leq n} = a_c + a_{c+1} + a_{c+2} + \cdots + a_n.$$

Series

Example 2.4: Product Notation

Determine the following products.

1. $\prod_{i=1}^{5} i$
2. $\prod_{i=0}^{6} (a_i)$ whwere $(a_i) = 2^i$

Solution

Series

Example 2.5: Product Notation

Determine the following products.

1. $\prod_{i=1}^{5} i$
2. $\prod_{i=0}^{6} (a_i)$ whwere $(a_i) = 2^i$

Solution

1. $\prod_{i=1}^{5} i = 1 \times 2 \times 3 \times 4 \times 5 = 5! = 120$
2. $\prod_{i=0}^{6} (a_i) = 2^0 \times 2^1 \times 2^2 \times 2^3 \times 2^4 \times 2^5 \times 2^6 = 2^{21} = 2097152$

Series

- The more general series describes a summation with a possibly infinite number of terms.
- Some authors use series and summation interchangeably in the finite case and instead say infinite series to emphasize that the series in question has infinitely many terms.
- Some special infinite series are said to be convergent if their sum approaches a fixed value in its limit.
- Otherwise, the series is said to be divergent.

Series

Example 2.6: Converging Series

The Series $\sum_{i=0}^{\infty} \frac{1}{2^n}$ is

$$\sum_{i=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

The partial sums of this series are defined as $\sum_{i=0}^{n} = 1 + \frac{2^n - 1}{2^n}$

Clearly taking $\lim\limits_{n\to\infty}$ we have

$$\sum_{i=0}^{\infty} \frac{1}{2^n} = \lim_{n\to\infty} 1 + \frac{2^n - 1}{2^n} = 2$$

Closed-form formulas for Summations

- Finding the closed-from of a summation entails finding a function f such that $f(n) = \sum_{i=0}^{n} a_i$ assuming the sequence (a_i) is defined for $i \in \mathbb{N}$
- The easiest closed form solution comes directly from the definition of multiplication.
- Indeed, multiplication is just repeated addition. For any constant number c we have the following summation formula.

$$\sum_{i=1}^{n} c = nc. \qquad \sum_{i=0}^{n} c = (n+1)c.$$

- One fundamental closed-form sum is:

$$\sum_{i=0}^{n} i = \frac{n(n+1)}{2}.$$

Closed-form formulas for Summations

Similarly, there are closed forms for $\sum_{i=0}^{n} i^2$ and $\sum_{i=0}^{n} i^3$

$$\sum_{i=0}^{n} i^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$$

$$= \frac{(n)(n+1)(2n+1)}{6}$$

$$\sum_{i=0}^{n} i^3 = \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}$$

$$= \frac{n^2(n+1)^2}{4}$$

Exercises

Example 2.7

Determine the sum of the following summation $\sum_{i=0}^{15} i^2 + 3i$

$$\sum_{i=0}^{15} i^2 + 3i = \sum_{i=0}^{15} i^2 + 3\sum_{i=0}^{15} i$$

$$= \frac{(15)(16)(31)}{6} + \frac{3}{2}(15)(16)$$

$$= 1600$$

Exercises

1. Determine the sum of the following summations

   a.) $\sum_{k=0}^{3} 3^k$

   b.) $\sum_{t=1}^{1} cos(\pi t)$

   c.) $\sum_{\ell=0}^{10} 6^2$

   d.) $\sum_{i=-2}^{2} 2i + 4$

2. Consider the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 4$. What are the first 7 terms of this sequence? Can you find a closed-form formula to a_n?

Exercises

3. Consider the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$. Are the following valid solutions to this recurrence relation? If so, compute $a_0, a_1, \ldots, a_4$ using the formula
   a.) $(a_n) = 0$
   b.) $(a_n) = n4^n$
   c.) $(a_n) = n^2 4^n$
   d.) $(a_n) = (2 + 3n)(4^n)$
4. Give a recurrence relation and initial condition(s) for:
   a.) The arithmetic progression $(c, c+s, c+2s, c+3s, \ldots)$
   b.) The Geometric progression $(bn, bn^2, bn^3, bn^4, \ldots)$

This is the End of
Discrete Structure 1

I wish you the best in
your examinations

Exam will be restricted
to the
Exercises and Examples